Machine Learning System Architecture
01 Introduction and Overview
02 Introduction and Overview
03 Feature Extraction
04 Feature Engineering
05 Feature Engineering Continued
06 Feature Selection Approach
07 Feature Selection Approach
08 Feature Selection Approach
09 Feature Selection Approach
10 Feature Selection Approach
11 Feature Selection Approach
12 Feature Selection Approach
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01
Introduction and Overview

We have written this book because we want to help machine learning (ML) engineers and data scientists to succeed at the system design interviews. This book can also be helpful for anyone who wants to improve their system design skills or who is just curious about such topics.

Many engineers think of ML algorithms when they hear about machine learning (ML) interviews such as the entirety of an ML system. However, an ML system is production ready much more than just model development. ML systems are typically complex, consisting of a variety of components and processes. Many of these components and processes are not ML algorithms and are not even related to ML, such as data ingestion, data transformation, and the deployment of the trained ML algorithm and parameters, and monitoring to ensure the system's performance doesn't degrade over time.

ML System

Figure 1.1 Components of a conventional ML system

At an ML system design interview, you are expected to answer open-ended questions. For example, you might be asked to design a route recommendation system or a photo sharing system. There is not enough time in a typical interview session. This interviewer wants to assess your thought process, your in-depth understanding of various ML topics, your ability to design an end-to-end system, and your ability to communicate clearly and make pros and cons of various options.

To succeed in designing complex ML systems, it is very important to follow a framework. Unstructured answers make the flow difficult to follow and can easily miss critical components. The framework we use in what throughout this book to resolve questions about ML system design. This framework consists of the following key steps:

1. Clarifying requirements
2. Frame the problem as an ML task
3. Data preparation
4. Model development
5. Evaluation
6. Deployment and serving
7. Monitoring and infrastructure

Figure 1.2 The framework we use to approach ML system design

Every ML system design interview is different because the interviewer is open-ended, and there's no one-size-fits-all way to succeed. The framework is intended to help you navigate the interview and ensure you don't miss critical components. It is flexible. It can intertwine is nearly interwoven in broader development, you should always adhere to the flow.

Let's now go into greater detail on each of the framework.

Clarifying Requirements

ML system design questions are usually intentionally vague, with the bare minimum of information. For example, an interviewer might ask you to design a video recommendation system. Your first step should be to ask clarifying questions to narrow the scope of the question. A solid plan, one should ask questions to understand the exact requirements. Some of the questions you should ask include:

- **Business objective**: If we are asked to create a system to recommend vacation rentals, two possible motivations are to increase revenue or to increase customer satisfaction.
- **Features the system needs to support**: What are some of the features that the system is expected to support which could affect our ML system design? For example, will users be able to search for specific rental properties or only be shown recommendations? Can users save their favorite rentals? Can users see their previously viewed rentals?
- **Data**: What are the data sources? How large is the dataset? Is data labeled?
- **Constraints**: How much computing power is available? Is it cloud based, or should the system work on a device? Is the model expected to have latency requirements? Is it memory constrained?
- **Scale**: How many users are there? How many items, such as videos, are we dealing with? What's the rate of growth of these metrics?
- **Performance**: How fast should predictions be? Is real-time solution needed? Does accuracy have more priority or latency?

This list is not exhaustive, but you can use it as a starting point. Remember that other factors, such as privacy and ethics, can also be important. At the end of the day, we're expected to reach an agreement with the interviewer about the scope and requirements of the system we are trying to design. Let's now look at some examples of how the list of requirements and constraints can affect the design of a system.

Frame the Problem as an ML Task

Business objectives often are not phrased in a way that is conducive to ML. Suppose an interviewer asks you to increase the user engagement on a video sharing platform. Is this a problem that can be solved with ML? Not necessarily. It could be that the best way to achieve this is by hiring more UI/UX designers to improve the app's design. Or it could be that the best way to increase engagement is by adding more features. Or it could be that the best way to increase engagement is by adding more content. Or it could be that the best way to increase engagement is by improving the app's recommendation system. Only the last option is a problem that can be solved with ML.

- **Business objective**: Increase user engagement on our video sharing platform
- **Features the system needs to support**: What are some of the features that the system is expected to support which could affect our ML system design? For example, will users be able to search for specific videos? Can users save their favorite videos? Can users see their previously viewed videos? Can users share videos with their friends? Can users see what their friends are watching?
- **Data**: What are the data sources? How large is the dataset? Is data labeled?
- **Constraints**: How much computing power is available? Is it cloud based, or should the system work on a device? Is the model expected to have latency requirements? Is it memory constrained?
- **Scale**: How many users are there? How many videos are there? How many users are dealing with? What's the rate of growth of these metrics?
- **Performance**: How fast should predictions be? Is real-time solution needed? Does accuracy have more priority or latency?

This list is not exhaustive, but you can use it as a starting point. Remember that other factors, such as privacy and ethics, can also be important. At the end of the day, we're expected to reach an agreement with the interviewer about the scope and requirements of the system we are trying to design. Let's now look at some examples of how the list of requirements and constraints can affect the design of a system.

Frame the Problem as an ML Task

Business objectives often are not phrased in a way that is conducive to ML. Suppose an interviewer asks you to increase user engagement on a video sharing platform. Is this a problem that can be solved with ML? Not necessarily. It could be that the best way to achieve this is by hiring more UI/UX designers to improve the app's design. Or it could be that the best way to increase engagement is by adding more features. Or it could be that the best way to increase engagement is by adding more content. Or it could be that the best way to increase engagement is by improving the app's recommendation system. Only the last option is a problem that can be solved with ML.

Application	Business objective
Ecommerce marketplace	Increase user engagement
Video streaming app	Increase user engagement
Ad click prediction system	Increase user clicks
Harmful content detection in a social media platform	Improve the platform's safety
Friend recommendation system	Increase the rate at which users grow their network

Table 1.1 Examples for business objectives and an ML objective

Specifying the system's input and output

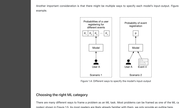
Once we decide on the ML objective, we need to define the system's inputs and outputs. For example, for a harmful content detection system we need a model that takes in a post and outputs a probability that the post is considered harmful or not.

Application	ML objective
Ecommerce marketplace	Increase the number of users registrations
Video streaming app	Maximize the time spent watching videos
Ad click prediction system	Maximize click through rate
Harmful content detection in a social media platform	Accurately predict if a given content is harmful
Friend recommendation system	Maximize the number of friend connections

Table 1.2 Examples for business objectives and an ML objective

Specifying the system's input and output

Once we decide on the ML objective, we need to define the system's inputs and outputs. For example, for a harmful content detection system we need a model that takes in a post and outputs a probability that the post is considered harmful or not.


Harmful Content Detection System
Probability of Harmful
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Semi-supervised Learning
Self-supervised Learning
Active Learning
Multi-task Learning
Ensemble Learning
Transfer Learning
Federated Learning
Meta Learning
Reinforcement Learning
Multi-task Learning
Ensemble Learning
Transfer Learning
Federated Learning
Meta Learning
Active Learning
Self-supervised Learning
Semi-supervised Learning
Unsupervised Learning
Supervised

Collect the raw data

This is extremely dependent on the data preparation step, so we do not repeat it here.

Identify features and labels

During the feature engineering step, we have already discussed which features to use. So, we'll focus on creating labels for this data. There are four common ways to get labels: hand-labeling and natural labeling.

Hand-labeling. This means hiring human contractors (read the only human in the process) to manually annotate or label whether a post contains information or not. Hand labeling leads to accurate labels since a human is involved in the process. However, hand-labeling is time-consuming, and it can be expensive if subject matter experts are required for labeling, such as in the case of medical records.

Natural labeling. In natural labeling, the ground truth labels are inferred automatically without human annotators. Let's use an example to better understand what this means.

Suppose we want to average all N Uber's ETA values for each hour, so we can plot a time series. One possible way to solve this task is to train a model to predict the ETA given the current time, the rider's location, and the driver's location. Then, for any ride, we can use the ride's start time as a training example, the ride's features as the input, and the ride's actual ETA as the label. For example, if a rider requested a ride at 2:47:21 PM, we can safely round down to the nearest hour, and use the ETA value we calculated for 2:00:00 PM as the label.

In this way, it is incredibly easy to create a large dataset, because we obtain training labels each time a driver completes a ride.

Select a sampling strategy

It is a common practice to collect all the data, or randomly or stratifiedly sample it to reduce the amount of data in the system. Common sampling methods include random sampling, stratified sampling, weighted sampling, reservoir sampling, and importance sampling. To learn more about sampling methods, you can refer to 31.

Split the data

It is the system's responsibility to split the processing of the dataset into training, validation (optional), and test datasets. To learn more about data splitting techniques, refer to 32.

Address class imbalances

A dataset with skewed class labels is called an imbalanced dataset. The class that makes up a larger proportion of the dataset is called the majority class, and the class that makes up a smaller proportion of the dataset is called the minority class.

Figure 7.6: Imbalanced dataset with two classes

An imbalanced dataset is a serious issue in model training, and it creates a model may not have enough data to learn the minority class. For example, if a dataset contains 1 million rows, it's fine if only 1 or 2 of them are minority class. Many techniques can be used to handle training rows and class imbalance.

Resampling training data

Resampling refers to the process of adjusting the ratio between different classes, making the data more balanced. For example, we can oversample the minority class (Figure 7.7) or undersample the majority class (Figure 7.8).

Figure 7.7: Oversampling the minority class

Figure 7.8: Undersampling the majority class

Attaching time functions

This technique allows the loss function to make a more robust against class imbalance. The high level idea is to give more weight to rare examples during loss calculation. Adding class weights to the loss function penalizes the model more when it makes a wrong prediction about a minority class. This forces the model to learn to correctly classify minority classes more effectively. This commonly used loss functions include weighted cross-entropy and class-balanced loss (9) and focal loss (10).

Choosing the loss function

For most common tasks/losses, we need to choose a proper loss function to train the model. A loss function is a measurement of how good a model is at predicting the expected outcomes. The loss function is used to calculate gradients and update the model's parameters during the training process in order to minimize the loss.

Dropping a row may be trivial in our case, it is OK. Alternatively, you can simply suppose it wasn't a rare function from a list of existing rare functions based on how you handle this problem. Nonetheless, you might need to make minor changes to the loss function to adapt to this new problem. We will provide more details about loss functions in Step 7.

Training from scratch vs. fine-tuning

Training a model from scratch means the model's training from scratch vs. fine-tuning. Fine-tuning means continuing to train the model on new data to adapt the model's weights to its target performance. This is a design decision you may need to discuss with the interviewer.

Distributed training

Training at scale becomes increasingly important because models grow bigger over time, and the size of the dataset also increases dramatically. Distributed training is commonly used to train the model. Distributed training can be done in various ways. Techniques such as data parallelism and model parallelism are commonly used to speed up model training. There are two main types of distributed training: vertical scaling and horizontal scaling.

Depending on what kind of ML problem, distributed training distributed training may be necessary. For example, in the Google Search team, we have to deal with the interview. Note that distributed training is a generic topic you can talk about irrespective of the specific task.

Tuning hyperparameters

Some tuning points are listed below:

- **Model architecture**: What ML models are suitable for the task, and what are their pros and cons, mainly a list of topics to consider during model selection:
  - The nature of the task
  - The amount of training data the model expects
  - The computing resources the model may need
  - The inference latency requirement
  - Can the model be distributed or is it a black-box?
  - Model's compatibility, relating to model-serve complexity may increase the performance, but the results might be harder to reproduce.
  - Can we leverage pre-training or transfer learning to decrease data hunger and speed?
  - How many parameters does the model have? How much memory is needed?
  - Is the model's output a single value, a sequence, or a multi-dimensional tensor? Are Bernoulli or Transformer based decoders needed? Do we need to use a pre-trained embedding? Do we need to use a pre-trained encoder? Do we need to use a pre-trained feature extraction, such as ResNet?
- **Dataset mixes**: How should we combine the dataset? Is data unbalanced, and if so, how good are the proportions? If natural labels are available, how do we get them? How do we receive user feedback on the system? How long does it take to get user feedback?
- **Model training**:
  - What loss function should we choose? e.g., Cross-entropy [19], MSE [20], MAE [20], Hinge loss [21], etc.?
  - Should we use regularization? such as L1 [18], L2 [18], Dropout [22], Stochastic Depth [23], MixUp [24], or Cutout [25]?
  - What is the optimizer? Adam?
  - You may need to describe common optimization methods such as SGD [26], Adam[27] [28], Nesterov[29], [30], and Adagrad [31].
  - Should we use a fixed learning rate or use an adaptive one? e.g., SGD, Cyclic [32], SGDR [33], Sparsed [34].
  - How to handle an imbalanced dataset?
  - How to initialize the model weights?
  - What are the activations? Sigmoid? ReLU? Leaky ReLU? ELU [35], SELU [36], Swish [37], Sparsed [38]?
- **Continual learning**: Do we want to train the model online with each new data point? Do we need to personalize the model to every user? How often do we retrain the model? Some models need to be retrained daily or weekly, and others monthly or yearly.

Evaluation

The next step after the model development is evaluation, which is the process of using different metrics to understand an ML model's performance. In this section, we examine two evaluation methods, offline and online.

Offline evaluation

Offline evaluation refers to evaluating the performance of ML models during the model development phase. To evaluate a model, we need to define evaluation metrics. Table 7.4 lists some of the common metrics for different tasks. Note that there are many other metrics for specific tasks. Table 7.4 lists some of the commonly used metrics for different tasks.

Task	Offline metrics
Classification	Precision, recall, F-score, accuracy, ROC, AUC, PR-AUC, confusion matrix
Regression	MSE, RMS, RMSE
Ranking	Precision@k, mAP@k, MRR, nDCG
Image generation	Inception score [39], Frechet inception distance [40]
Natural language processing	BLEU [20], METEOR [41], ROUGE [42], CIDEr [43], SPICE [44]

Table 7.4: Popular metrics in offline evaluation

During an interview, it's important to identify business-relevant metrics. Discuss the pros and cons of each and how we have tradeoffs. For example, if the task is a ranking problem, an engineer should know the ranking metrics such as mAP and MRR.

Online evaluation

Online evaluation refers to the process of evaluating how the model performs in production after deployment. To measure the impact of the model, we need to define different metrics. Online metrics are close to our goals, and we can directly or with indirection and are usually tied to business objectives. Table 7.5 shows various metrics for different problems.

Problem	Online metrics
Ad click prediction	Click through rate, revenue lift, etc.
Harmful content detection	Prevalence, reshare rate, etc.
Video recommendation	Watch time, watch time per number of completed videos, etc.
Friend recommendation	Number of requests sent per day, number of requests accepted per day, etc.

Table 7.5: Metrics for various problems

In practice, companies usually track many online metrics, or an alternative, we need to select some of the most important ones to measure the impact of the system. It is important to be able to affirm relevance, interpret the metrics, and be able to explain. Metrics should be sensitive to small changes and business stakeholders.

In the step, the interviewer gauges your business sense. So, it's helpful to communicate your thought process and your lessons learned from choosing certain metrics.

Talking points

Some are some talking points for the evaluation step:

- Offline metrics are important for measuring the effectiveness of the ML system online? How do these metrics relate to the business objective?
- Do offline metrics correlate well with the online metrics? Why or why not? Discuss the pros and cons of each.
- Fairness and bias: Are the metrics good enough for various different subgroups, such as age, gender, race, etc.? How should we fix? What happens if someone with malicious intent gains access to your system?

Deployment and Serving

A natural next step after choosing the appropriate metrics for online and offline evaluations is to deploy the model to production and serve millions of users. Some important topics to cover include:

- Cloud vs. on-device deployment
- Model compression
- Testing in production
- Prediction pipeline

Let's take a look at each.

Cloud vs. on-device deployment

Deploying a model on the cloud is different from deploying it on a mobile device. Table 7.6 summarizes the main differences between on-device and cloud deployment.

	Cloud	On-device
Simplicity	✓ Simple to deploy and manage using cloud-based services	✗ Deploying models on a device is not straightforward
Cost	✗ Cloud costs might be high	✓ No cloud cost when computations are performed on device
Network latency	✗ Network latency is present	✓ No network latency
Inference latency	✓ Usually faster inference due to more powerful machines	✗ Will, mostly, be slower
Hardware constraints	✓ Fewer constraints	✗ More constraints, such as limited memory, battery consumption, etc.
Privacy	✗ Data privacy is a concern because it leaves the device	✓ Raw privacy-sensitive data never leaves the device
Dependency on internet connection	✗ Internet connection needed to send and receive predictions	✓ No internet connection needed

Table 7.6: Trade-offs between cloud and on-device deployment

Model compression

Model compression refers to the process of making a model smaller. This is necessary to reduce the inference latency and model size. Some techniques commonly used to compress models include:

- **Knowledge distillation**: The goal of knowledge distillation is to train a small model (student) to mimic a larger model (teacher).
- **Pruning**: Pruning refers to the process of finding the lesser useful parameters and setting them to zero. This leads to sparsed models that can be compressed efficiently.
- **Quantization**: Most parameters are often represented with 32-bit floating-point numbers in a typical ML system. Quantization is an approach to represent the parameters, which reduces the model's size. Quantization can happen during training or post-training (31).

To learn more about model compression, you can refer to 32 and 33.

Test in production

The only way to ensure the model will perform well in production is to test it with real traffic. Commonly used techniques to test models include shadow deployment [34], A/B testing [35], canary release [36], interleaving experiments [37], bandits [38], etc.

To demonstrate that we understand how to test in production, it's a good idea to mention at least one of these techniques. Let's briefly review shadow deployment and A/B testing.

Shadow deployment

In this method, we deploy the new model in parallel with the existing model. Each incoming request is routed to both models, but only the existing model's prediction is served to the user.

By shadow deploying the model, we minimize the risk of unknown predictions and the newly developed model has been thoroughly tested. However, this is a costly approach that requires additional computing resources.

Figure 7.9: Shadow deployment

A/B testing

With this method, we deploy the new model in parallel with the existing model. A portion of the traffic is routed to the newly developed model, while the remaining requests are routed to the existing model.

In order to execute A/B testing correctly, there are two important factors to consider. First, the traffic routed to each model has to be random. Second, sight-bias should be avoided. For example, if we conduct a product test, we want the users to be oblivious

Figure 7.10: A/B testing

Prediction pipeline

To serve frequent in production, we need a prediction pipeline. An important design decision to make is the choice between online prediction and batch prediction.

Batch prediction. When batch prediction, the model makes predictions periodically. Batch prediction jobs are not computed, so don't need to worry about how long it takes the model to generate predictions since they are computed.

However, the batch prediction approach has several drawbacks. First, the model becomes less responsive to the changing preferences of users. Secondly, batch prediction is only possible if we know in advance what needs to be pre-computed. For example, in a recommendation system, we can pre-compute a user's recommendations in advance and retrieve them when needed.

Figure 7.11: Batch prediction workflow

Online prediction. In online prediction, predictions are generated and returned as soon as requests arrive. The main problem with online prediction is that the model takes more time to generate predictions.

Figure 7.12: Online prediction workflow

This choice of batch prediction or online prediction is mainly driven by product requirements. Online prediction is generally preferred if situations where we do not know what needs to be computed in advance. Batch prediction is ideal when the system recommends items to users that do not rapidly change based on user preferences.

In practicals discussions, ML system design interviews more than just the modeling. Engineering is overall the system design in an interview demonstrates a deep understanding of how different components work together as a whole. Interviewers often take this as a critical signal.

Figure 7.13 shows an example of the ML system design for a recommendation system. We will examine it in more depth in Chapter 9.

Figure 7.13: An system design for a personalized news feed system

Talking points

- Is a model computation needed? What are some commonly used computation techniques?
- Is a time prediction or batch prediction more suitable? What are the trade-offs?
- Do we need to use a feature store? Why or why not?
- How should we deploy the developed model in production?
- For ML system consists of various components (working together to serve requests. What are the responsibilities of each component? Why?
- What technologies should we use to ensure that serving is fast and scalable?

Monitoring

Monitoring refers to the task of tracking, measuring, and logging different metrics to ensure ML system in production can for many reasons. Monitoring may be important because these metrics can tell us when there are flaws in the system.

Monitoring is an important topic to discuss in ML system design interviews. You primary where you may want to discuss are:

- Model staleness
- Monitoring metrics

Why a system fails in production

There are various reasons why an ML system may fail after it is deployed to production. One of the most common reasons is data distribution shifts.

Data distribution shift refers to the phenomenon in which the data a model receives in production differs from that it encountered during training. Figure 7.14 shows an example where the training data consisted of cages users from three sizes, but an serving time, an image of a cat in a different position is passed to the ML model.

Figure 7.14: Data distribution shift

Data distribution in the real world commonly changes, and it's important to monitor it. Monitoring is likely to become less relevant over time process. The model is trained on data that stationary performance over time. Monitoring data distribution changes is important because it allows us to know when to update the model. For example, if we know that the data distribution changes every month, we can update our model to learn this problem. Some commonly approaches for dealing with data distribution shifts include:

- Collecting training data from the new distribution
- Adding more data from the old distribution
- Adaptive machine learning algorithms

Operation-related metrics. Production models should be treated as and similar. They include average serving time, throughput, the number of prediction requests, CPU/GPU utilization, etc.

ML-specific metrics

- Monitoring hyperparameters. Models are only as good as the data they consume, so monitoring the model's input and outputs is critical.
- Online, input to a deployed model to predict values are monitored to detect changes in their underlying distribution.
- Model accuracy. For example, be expect the accuracy to be within a specific range.
- Model variance. Model which version of the model is deployed.

Infrastructure

Infrastructure is the foundation for testing, deploying, and maintaining ML systems.

In every ML interviews, you can't be asked infrastructure related questions. However, some ML roles, such as DevOps and SE-ML roles require infrastructure knowledge. So it's important to be clear about the interviewer's expectations for the role.

Infrastructure is a very broad subject and cannot be summarized in just a few pages. So, it you're interested in learning more about ML infrastructure, refer to 14[39][40].

Summary

In this section, we proposed a framework for ML system design interviews. While many topics discussed in this chapter are not specific to ML problems and applicable to a wide range of tasks. Throughout the book, we will focus on unique talking points specific to the problem and ways to solve each exercise. For example, data preprocessing, feature engineering, and model infrastructure are often similar, regardless of the task. Therefore, we do not repeat generic topics in later chapters, but we will discuss unique talking points for each problem.

Finally, to engineer and design a system that meets the needs of the ML lifecycle, from business requirements development and product sense, while others specialists in model development. Many ML system design interviews are open-ended interviews, while others may focus heavily on a specific area, such as model development. Regardless of the role and the interviewer's preference, some steps may be skipped or scoped down or expanded upon. Depending on the role and the interviewer's preference, some steps may be discussed in more detail, while others may be discussed briefly or even skipped. In general, a candidate should seek to drive the conversation, while being open to the interviewer's feedback and hints. Table 7.7 summarizes the main points.

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